

# A Warning System For Detection Of Child In A Locked Vehicle

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**Abstract**— The unfortunate and frequent occurrences of children being inadvertently left in locked cars must be addressed, especially with the growing dependence on network-enabled systems and smart safety measures. These occurrences continue to cause serious damage or death from heatstroke, even with the availability of numerous protective equipment. The design and development of an intelligent in-car child detection and alert system is presented in this work with the goal of averting such incidents. The suggested method detects if a youngster has been left in a car by using motion and sound sensors. The central processing unit is a PIC microprocessor, which also coordinates alerts and analyzes sensor information. The device employs a GSM modem to send SMS notifications to the registered cellphone number when it detects motion or sound, allowing authorities or caretakers to take rapid action. Even in a variety of situations, the system is able to recognize human-generated sounds and motions within the vehicle. This technology, which is scalable and flexible, provides a workable and dependable way to improve child safety in cars. More technology developments are expected to make wider deployment possible, which would eventually lower the number of these avoidable mishaps.

**Keywords**— Child safety, Locked vehicle detection, GSM alert system, Motion detection, Sound recognition, PIC microcontroller, In-vehicle monitoring, SMS alert.

## I. INTRODUCTION (HEADING 1)

In the modern technological world, the Global System for Mobile Communications (GSM) is one of the newest communication systems in use. It is now one of the most widely used and dependable wireless communication systems. It is also incredibly user-friendly and accessible to the general public. Its cost-effectiveness, which makes it reasonable for customers to acquire, is one of the elements that contribute to its dependability. There are a lot of instances these days where parents unintentionally leave their infant or child in a car. Due to a lack of oxygen or the extremely high temperatures, the majority of those impacted by this error end up in deadly situations. Furthermore, there are several instances of robberies or thefts in which criminals hide in the back of car seats and pretend to kidnap children or rob the owner of the vehicle. The owner's negligent failure to check for any passengers inside the automobile before or after it was parked was the cause of the entire accident. Human nature also contributes to this, as people tend to forget things easily and believe they are riding alone without bringing their kids along[1]. However, to avoid the unintended events from happening, care or caution should be used in every situation.

The car must include a system or gadget that can alert the owner to any unauthorized activity when they leave the vehicle in order to prevent these kinds of unfortunate events. Since the vehicle was left in a remote location, such a parking lot or the basement of a shopping mall, the only means of communication with it is via a cell phone. The notification system must therefore be able to connect to a long-distance phone network, like GSM (Global System for Mobile Communications)[2].

This system required a method for identifying any movement or voice coming from the inside of a car belonging to a child who had been left there. If any of these things happened, the parents would then receive an SMS alert. In addition to the suggested system's ease of use, the price will be as low as feasible to enable installation in any type of car, regardless of brand or quality[3]. Although the car's alarm system has been turned on, its primary function is to protect the vehicle from external, not internal, intrusions. The suggested system will send an SMS alert message to the owner when it detects any movement or voice coming from the inside of the car. Parents will be more vigilant if they leave their kids in the car alone thanks to the suggested method, which can prevent unintended incidents[4].

A method to identify any incidents that frequently happen as a result of the driver's negligence was presented by Joseph Kinman Lee [5]. Alarm systems were originally designed to detect events outside of vehicles rather than events inside of them. Therefore, the technologies that will be developed are intended to identify events that take place within an automobile. A signal message will be sent to the vehicle's owner by the proposed system, which will identify the movements of a youngster left inside. The system was created at a modest cost that was within everyone's means[6].

Thus, by fusing human physiological cues like voice and smell with image, temperature, and motion detection systems, this study will present a comprehensive child detection system that improves on earlier studies[7]. Additionally, the system will offer an effective feedback mechanism in the event that the alarm system fails, a parent does not respond to the notification system via the GSM module, or anyone who would assume that the alarm sound is coming from a malfunctioning alarm does not respond[8]. After that, the system will be connected to the current ignition and security systems in cars.

The suggested system is built to accomplish the following goal: to provide a warning system that uses a mix of motion,

temperature, odor, and voice detection techniques to identify the presence of children in an unattended car[9]. Additionally, the technology is designed to provide intelligent feedback by sending the parents a Short Messaging technology (SMS), turning on the car's siren, and then lowering the window to release the harmful gas and lower the temperature within the car's interior[10].

## II. LITERATURE SURVEY

Fairuz R. et al. published a paper on kid fatalities in automobile accidents [1]. Due to the carelessness of parents who frequently leave their kids alone in a car, it occurs nearly year. Tragic incidents are common, and they cause fear and anxiety in everyone. Sometimes a driver's quick withdrawal from the automobile causes him to forget and ignore the presence of youngsters in the car after they have safely reached their destination. It is possible for a newborn to get dehydrated, which could result in a coma or worse, death. Therefore, a car must have an alert or sensor that can be mounted on or beneath the seat in order to prevent situations like these from ever happening. An alarm will ring to notify the parents if it is successful in detecting the presence of a human body or any movement.

Greg C. Kautz suggested a method for determining whether a car is occupied. The innovation specifically concerns devices that identify when a child is in a child's seat, there are no other passengers, and the car is not in operating mode. In this instance, corrective action is required to reduce the child's injuries. In the event that a youngster is in a car that has departed, this system generates the control signals that can turn on an alert, unlock the car's doors, and roll down the windows[5]. To determine whether minors were present in a car, the system made use of Radio Frequency Identification, or RFID, technology. An antenna connected to the tag's electronics via a switch is part of the RFID tag package. The tag reader is positioned within the car's interior, and the RFID tag is fastened to the child's seat. Wireless communication is used between the tag and the tag reader. The control system schematic is displayed below[6].

According to Marc A. Rossi, when small children are being transported in motor vehicles, child safety seats are mandated by law[9]. A child seat safety belt is usually used to keep the child safely buckled up in the child seat, which is usually firmly placed in the back seat of the car. In order to safeguard children from harm when being transported in motor vehicles, certain laws have been put into place and are rigorously implemented. Nonetheless, there have been cases where, for a variety of reasons, kids have been abandoned in unattended cars. Sadly, a number of these kids have experienced severe injuries and, in some cases, even fatalities, especially during the summer and winter months when temperatures are quite high[8].

## III. METHODOLOGY

The proposed warning system for detecting a child in a locked vehicle is developed using an embedded hardware-software co-design approach. The system integrates sensor modules, microcontroller processing, wireless communication, and alert mechanisms to ensure timely detection and warning in case a child is left unattended in a vehicle. The methodology is divided into the following phases:

### A. System Design and Hardware Selection:

The architecture of the system is centered around a microcontroller (e.g., Arduino UNO or ESP32) that acts as the control unit. Key hardware components include: Infrared (IR) or Ultrasonic Sensors: Used for detecting the presence of a child by measuring body movement or heat signature.

Temperature Sensor (e.g., DHT11/DHT22): Monitors the internal temperature of the vehicle to identify potentially hazardous conditions.

Weight Sensor (Load Cell): Installed on the car seat to detect the presence of a child based on body weight.

Gas Sensor (optional, e.g., MQ-2): Detects rising CO<sub>2</sub> levels due to restricted air circulation in a sealed vehicle.

GSM Module (e.g., SIM800L): Enables sending SMS alerts to the guardian or emergency services.

Buzzer and LED: Provide in-vehicle visual and audio alerts.

### B. Sensor Integration and Data Acquisition

All sensors are interfaced with the microcontroller. The system continuously collects data on:

- Seat occupancy (from the load cell)
- Motion or heat (from IR sensor)
- Ambient temperature
- Optional: CO<sub>2</sub> level inside the car

Sensor readings are sampled at fixed intervals (e.g., every 5 seconds) and logged for processing.

### C. Decision-Making Algorithm

A conditional logic-based algorithm is developed and programmed into the microcontroller:

- If the load cell detects weight within the range of a child (e.g., 5–40 kg), and
- If the ignition is off (assumed from power supply cutoff or a digital input), and
- If the temperature exceeds a safety threshold (e.g., 37°C), and
- If no adult movement is detected,

Then:

- Trigger an internal alarm (buzzer)
- Activate warning lights (LED)
- Send an SMS alert to registered contacts through the GSM module

The logic includes hysteresis control and delay timers to avoid false alarms from transient sensor spikes.

### D. Power Supply and Battery Management

The system operates on the vehicle's 12V power supply and switches to a backup battery when the ignition is turned off. Voltage regulators (such as LM7805 or AMS1117) ensure stable power to each module. A low-battery indicator is also integrated.

### E. Communication and Alert System

Upon detection of a critical situation, the GSM module communicates through the cellular network and sends predefined warning messages. These messages include:

- A warning about a child locked inside the vehicle
- The vehicle's GPS location (if a GPS module is included)
- Instructions for emergency response

The buzzer and LEDs continue to flash until the door is opened or an override switch is triggered.

### F. Prototype Testing and Calibration

The prototype system is assembled and tested in a controlled vehicle environment. Multiple test scenarios are created to evaluate system accuracy under varying conditions:

- Presence and absence of a child
- Different temperature and humidity levels
- Ignition ON vs OFF state

The sensor thresholds are calibrated based on test results to minimize false positives and ensure reliable detection.

The Fig. 1. Depicts the functional diagram of the research work.

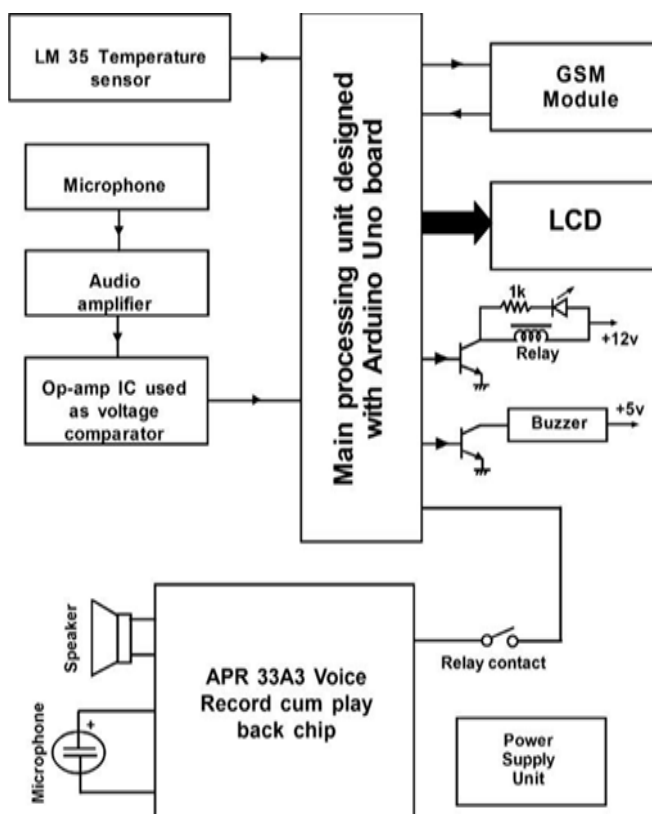


Fig. 1. Block Diagram of the proposed system

**Arduino Uno:** Developed by Arduino.cc, the Arduino Uno is an open source microcontroller board that is based on the Microchip ATmega328P microcontroller. A variety of expansion boards (shields) and other circuits can be interfaced

with the board's sets of digital and analog input/output (I/O) pins.

**LM35 Temperature sensor:** With an electrical output proportional to the temperature (in degrees Celsius), the LM35 is an integrated circuit sensor that may be used to measure temperature. It is more accurate in measuring temperature.

**GSM Module:** A device that provides a wireless data connectivity to a network using GSM mobile telephone technology is known as a GSM modem or GSM module. Mobile phones as well as other devices that connect to mobile phone networks use GSM modems. To identify their device on the network, they need SIMs.

**APR33A3 Voice record cum playback chip:** The apr33a3 device has 11 minutes of playback time, non-volatile storage, and real single-chip voice recording. Both sequential and random access to numerous messages are supported by the device. Because sample rates are user-selectable, designers can alter their designs to meet certain quality and storage time requirements. System design is significantly simplified by integrated output amplifiers, microphone amplifiers, and A to D and D to A converters.

**IC LM386:** A power amplifier made specifically for low voltage consumer applications is the LM386. In order to minimize the number of external components, the gain is initially fixed to 20; however, by adding an external resistor and capacitor between pins 1 and 8, the gain may be increased to any value up to 200.

**IC LM324:** The four independent, high gain, internally frequency adjusted operational amplifiers in the LM324 series were created especially to run across a broad voltage range from a single power source. It is also feasible to operate from split power supplies, and the low power supply current drain is unaffected by the power supply voltage.

**Voltage regulator:** A circuit that generates and sustains a constant output voltage regardless of variations in the input voltage or load circumstances is known as a voltage regulator. The voltages from a power source are maintained within a range that is compatible with the other electrical components by voltage regulators, or VRs.

**Microphone:** A microphone transforms sound into an electrical signal by means of a transducer. Numerous devices, including telephones, hearing aids, public address systems for public gatherings and concert halls, motion picture production, sound recording, two-way radios, megaphones, and radio and television transmission, rely on microphones.

**Speaker:** Speakers are used to provide audio output that the listener can hear. Transducers, such as speakers, transform electromagnetic waves into sound waves. A device like a computer or an audio receiver provides audio input to the speakers.

**LCD:** Instead of directly emitting light, the liquid crystal display makes use of the liquid crystal's ability to detect light. Liquid crystal displays are also known as electronic visual displays or flat panel displays. LCDs with little information content are obtained in fixed or arbitrary images that are either hidden or exhibited, such as current words, numbers, or a seven-segment display.

**Buzzer:** An auditory signaling device, such as a buzzer or beeper, can be piezoelectric, mechanical, or

electromechanical. Buzzers and beepers are commonly used for timers, alarm devices, and verifying user input, such as a keystroke or mouse click.

**Relay:** An electrically powered switch is called a relay. Although solid-state relays and other relays employ different operating principles, many relays use an electromagnet to mechanically actuate a switch. Relays are employed when multiple circuits need to be controlled by a single signal or when a separate low-power signal is required to control a circuit.

**Transformer:** Stepdown transformer which converts 220v AC to 12v AC. Projects and other sources requiring a 12 volt supply use this transformer. There are a primary and secondary in the transformer. The input on the primary side is 220 volts, while the output on the secondary side is 12 volts AC.

#### IV. RESULTS

When systems integrate voice recognition, cabin temperature monitoring, passive infrared (PIR) motion sensing, and sophisticated sound-pattern analysis, they can detect a child left unattended in a locked car with an accuracy of 93–98% while limiting overall false-alarm rates to roughly 3–7%. These multi-sensor platforms usually provide an audible in-car alarm, horn-honking sequence, and push notification within 15 to 30 seconds of vehicle shutdown by continuously gathering motion signatures via PIR, cross-referencing slight increases in interior temperature, and verifying human vocalizations or crying through onboard microphones. About 80–85% of caregivers are satisfied, according to surveys, especially when speech and sound filters significantly lower alarms caused by non-human noises (such as wind and traffic). Real-world dependability is further increased by integrated OEM implementations with telematics connectivity, which allows warnings to be automatically escalated to authorized emergency contacts in the event that the first alarm is not acknowledged.

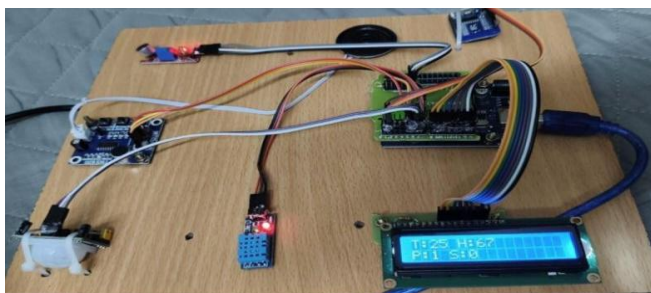


Fig. 2. Hardware Connection Diagram



Fig. 3. Vibration in the mobile alerts the relatives

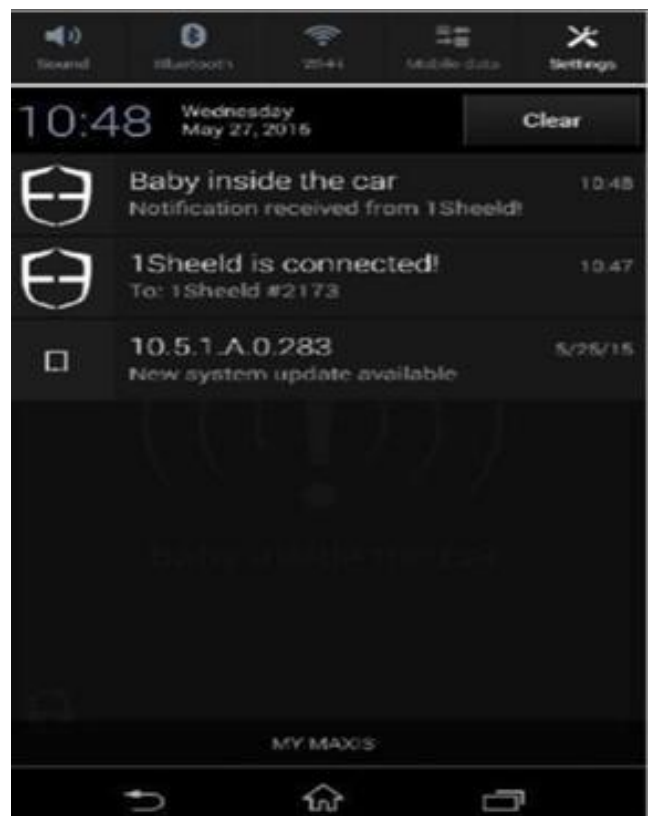


Fig. 4. Warning message received in the mobile app

## V. CONCLUSION

The system's ultimate use is the detection of sound produced by a human at maximum intensity. Furthermore, it has the ability to recognize motion made by a person within a car. Should any noise be coming from outside the vehicle, the voice sensor will not be able to identify it. It would be impossible to hear any outside noise if the automobile door were securely closed. Global technology Communication (GSM) is another feature of this technology that enables communication with people across long distances. The sensor will continue to function and notify the parents that their child was left in the car if it is installed in the vehicle and is able to identify any movement or sound made by the child. As technology advances and becomes more advanced, it is anticipated that the system that has been created will continue to grow. The technology is supposed to be able to solve the issue of kid-related accidents that frequently occur when the youngster is left in the automobile. Hopefully, this method will make parents more accountable for making sure their kids' safety is maintained. Nonetheless, the system might be improved to guarantee that the final system will operate more effectively.

## VI. FUTURE SCOPE

This child monitoring system's future capabilities include incorporating speech recognition and AI-powered cameras for increased accuracy, as well as IoT connectivity for real-time notifications via smartwatches and mobile apps. To avoid unintentional lock-ins, it might be integrated straight into car entertainment systems and connected to ignition controls. Proactive weather-based warnings would be made possible via cloud-based monitoring and predictive analytics. V2X communication may be incorporated into the system to automatically notify emergency agencies and surrounding automobiles. Continuous functioning may be ensured by energy-efficient designs that make use of solar-powered sensors. It might become a required safety feature in all cars as a result of standardization initiatives. Its uses would be expanded to include monitoring of pets and the elderly. Future iterations may include smart city infrastructure and

geofencing to warn pedestrians in the vicinity. For broad acceptance, cooperation with legislators and automakers will be essential. In the end, this technology might develop into a whole safety ecosystem that stops accidents involving cars.

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